Please check the examination details below before entering your candidate information

Candidate surname	Other names
Master Functional Skills	Model Solutions (MFS)

Pearson Edexcel Functional Skills

Centre Number	Candidate Number
MFS-UK	L2-P05

Past Paper 5

Time: 25 minutes | Paper Reference **PMAT2/N05**

Mathematics

Level 2
Section A (Non-Calculator)



You must have:
Pen, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks 16 /16

My signature confirms that I will not discuss the content of the test with anyone.

Signature: Master Functional Skills (MFS)

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Sign the declaration.
- Answer **all** questions.
- Write your final answers in the boxes provided.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You **must** show clearly how you get your answers in the spaces provided. Marks will be awarded for your working out.
- Check your working and answers at each stage.
- Diagram are **not** accurately drawn, unless otherwise indicated.
- **Calculators may not be used**.
- Take the value of π to be 3.14

Information

- The total mark for this section is 16.
- The marks for **each** question are shown in brackets
  – *use this as a guide as to how much time to spend on each question.*
- This sign ☑ shows where marks will be awarded for showing your checks.

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

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SECTION A

Answer ALL questions. Write your answers in the spaces provided.

1 Gareth sells hot drinks.
He sells cups of tea, cups of coffee and cups of hot chocolate.

Of all the hot drinks he sold last week

- $\frac{2}{5}$ were cups of tea
- $\frac{1}{3}$ were cups of coffee.

What fraction of all cups of hot drinks sold were cups of hot chocolate?

(3)

Step 1: Find the combined fraction of Tea and Coffee

Tea = $\frac{2}{5}$ Coffee = $\frac{1}{3}$

Find common denominator (LCM of 5 and 3 = 15):

$\frac{2}{5} \times \frac{3}{3} + \frac{1}{3} \times \frac{5}{5} = \frac{6}{15} + \frac{5}{15} = \frac{11}{15}$ [M1 - Common Denom]

Step 2: Subtract from 1 whole to find Hot Chocolate fraction

Hot Chocolate = $1 - \frac{11}{15} = \frac{15}{15} - \frac{11}{15} = \frac{4}{15}$ [M1 A1]

• 4 and 15 have no common factors, so 4/15 is already in simplest form.

$\frac{4}{15}$

(Total for Question 1 is 3 marks)

2 (a) Work out 13.4 × 5.2

(2)

Step 1: Multiply whole numbers ignoring decimal points (134 × 52):

```
  1  2     (carries for ×5)
  1 3 4
x   5 2
-------
  2 6 8     <- (134 × 2)
+ 6 7 0 0   <- (134 × 50)
-------
  6 9 6 8
```

Step 2: Position the decimal point:

- 13.4 has 1 decimal place
- 5.2 has 1 decimal place

Total = 1 + 1 = 2 decimal places -> 69.68 [M1 A1]

69.68

☑ (b) Use estimation to show a check of your working.

(1)

Round each number to a convenient whole number to check magnitude:

13.4 ≈ 13 5.2 ≈ 5

Estimated product = 13 × 5 = 65

- 65 is very close to 69.68, confirming the decimal point placement is correct.

(Alternatively: 10 × 5 = 50 or 15 × 5 = 75) [B1 - Estimation]

13 × 5 = 65

(Total for Question 2 is 3 marks)

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3 Mitch works in a fitness centre.
He gives one raffle ticket to each of 1000 people that attend the fitness centre.

Mitch starts to complete this two-way table to show information about the people he gives the raffle tickets to.

		favourite class			
		spin	**yoga**	**cardio**	**total**
age (years)	**18 to 24**	85	92	145	322
	25 to 49	107	119	134	360
	50 and over	38	170	110	318
	total	230	381	389	1000

(a) Complete the two-way table for Mitch.

(2)

Step-by-step table calculations:

- *18 to 24 spin: 322 − (92 + 145) = 322 − 237 = 85*
- *50+ cardio: 318 − (38 + 170) = 318 − 208 = 110*
- *25 to 49 yoga: 381 − (92 + 170) = 381 − 262 = 119*
- *25 to 49 total: 107 + 119 + 134 = 360 (Sum check: 1000)* *[M1 A1]*

Mitch selects a raffle ticket at random.

(b) What is the probability that the ticket he selects belongs to a person aged under 50 whose favourite class is cardio?

(2)

Step 1: Number of people aged under 50 with favourite class cardio:

Aged under 50 = (18 to 24) + (25 to 49)

Cardio under 50 = 145 + 134 = 279

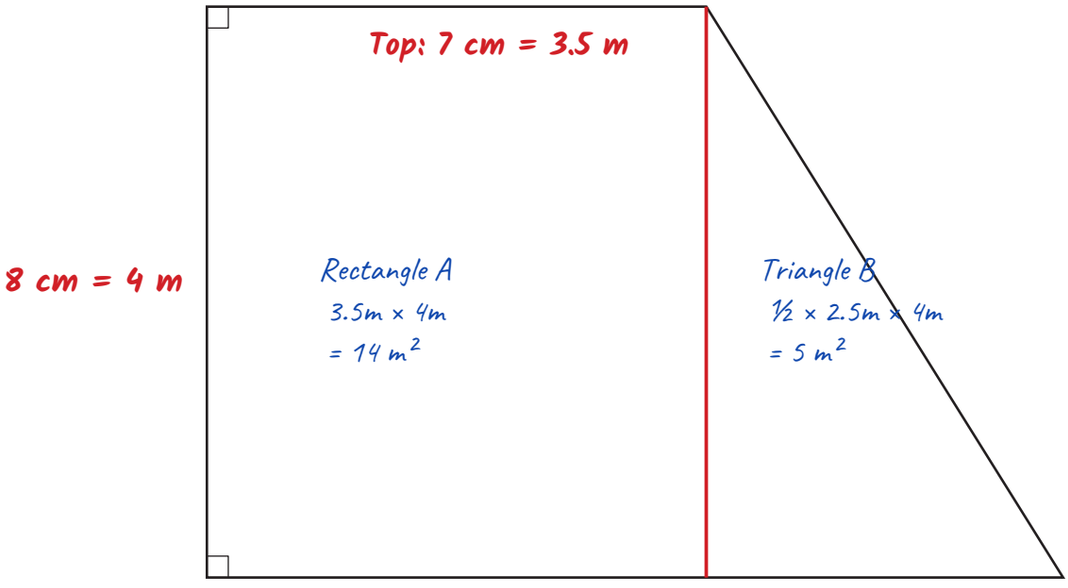
Step 2: Probability = 279 / 1000 = 0.279 (or 27.9%)

$\frac{279}{1000}$ (0.279)

[M1 A1]

(Total for Question 3 is 4 marks)

4 Dave is going to paint a wall.
Here is a scale diagram of the wall.



Scale 1 cm represents 0.5 m

Dave will paint this wall twice.
He thinks the total area that he needs to paint is greater than 30 m^2

Is Dave correct?
You **must** show all your working.

(6)

Step 1: Convert measurements using scale (1 cm = 0.5 m)

- Top = 7 cm × 0.5 = 3.5 m
- Height = 8 cm × 0.5 = 4 m
- Bottom = 12 cm × 0.5 = 6 m

[M1 M1 - Scale]

Step 2: Calculate Area of Wall (Trapezium or Rect + Tri)

- Area of Rectangle A = 3.5 × 4 = 14 m^2
- Area of Triangle B = ½ × (6 − 3.5) × 4 = ½ × 2.5 × 4 = 5 m^2
- **Total Wall Area = 14 + 5 = 19 m^2**

[M1 M1 - Area of Wall]

Step 3: Dave paints the wall TWICE

Total area to paint = 19 m^2 × 2 = 38 m^2

[M1 - Paint Twice]

Step 4: Comparison & Conclusion

38 m^2 is greater than 30 m^2 (38 > 30)

Yes, Dave is correct.

[A1 - Decision]

Decision: Yes, Dave is correct (38 m^2 > 30 m^2)

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Question 4 Full Solution & Verification Summary

Alternative Verification Method (Area scale factor):

1. Area on scale diagram = ½ × (7 + 12) × 8 = ½ × 19 × 8 = 76 cm^2
2. Since 1 cm represents 0.5 m, 1 cm^2 represents $(0.5)^2 = 0.25$ m^2
3. Real wall area = 76 × 0.25 = 19 m^2
4. **Painted twice = 19 × 2 = 38 m^2**
   -> 38 m^2 > 30 m^2 -> Dave is correct (Confirmed)

Specialist Tutor Tip for Area Scale Factor Questions:

- Common Trap: Never multiply final area by 0.5!
  Area scale factor is $(0.5)^2 = 0.25$.
- Best Practice: Convert every linear measurement to metres first
  (e.g. 7 cm -> 3.5 m, 8 cm -> 4 m, 12 cm -> 6 m).
- Always re-read carefully: Dave paints the wall twice (× 2)!
- Full marks require accurate figures (38 m^2 vs 30 m^2) and "Yes".

Official Pearson Edexcel Mark Scheme Grid (6 Marks):

- [M1] Process to interpret scale diagram measurements (7, 8, 12 cm)
- [M1] Process to apply scale factor 0.5 (3.5 m, 4 m, 6 m or 0.25 m^2/cm^2)
- [M2] Full process to calculate wall area (19 m^2)
- [M1] Process to double the area: 19 × 2 = 38 m^2

Yes (38 m^2 > 30 m^2) ✓

(Total for Question 4 is 6 marks)

TOTAL FOR SECTION A IS 16 MARKS

Please check the examination details below before entering your candidate information

Candidate surname	Other names
Master Functional Skills	Model Solutions (MFS)

Pearson Edexcel Functional Skills

Centre Number	Candidate Number
MFS-UK	L2-P05

Past Paper 5

Time: 1 hour 30 minutes | Paper Reference **PMAT2/C05**

Mathematics

Level 2
Section B (Calculator)



You must have:
Pen, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks: 48 /48

My signature confirms that I will not discuss the content of the test with anyone.

Signature: Master Functional Skills (MFS)

Instructions

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- Answer **all** questions.
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- Answer the questions in the spaces provided – *there may be more space than you need.*
- You **must** show clearly how you get your answers in the spaces provided. Marks will be awarded for your working out.
- Check your working and answers at each stage.
- Diagram are **not** accurately drawn, unless otherwise indicated.
- **Calculators may be used**.
- If your calculator does not have a π button take the value of π to be 3.14

Information

- The total mark for this section is 48.
- The total mark for this paper is 64.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*
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Advice

- Read each question carefully before you start to answer it.
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SECTION B

Answer ALL questions. Write your answers in the spaces provided.

1 Here is a set of data.

15 10 16 8 14 0 13 8 15 8

(a) Find the median.

(2)

Step 1: Order the 10 data values in ascending order:

0, 8, 8, 8, 10, 13, 14, 15, 15, 16

^ middle pair (5th & 6th values)

Step 2: Calculate the mean of the two middle values:

$$\text{Median} = \frac{10 + 13}{2} = \frac{23}{2} = 11.5$$

[M1 A1]

11.5

(b) Find the mode.

(1)

The mode is the most frequently occurring value in the data set.

- 8 appears 3 times
- 15 appears 2 times; all other numbers appear once

Mode = 8

[B1 - Mode]

8

(Total for Question 1 is 3 marks)

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2 Last year Jim bought 15 prizes for work.
The prizes cost £27 each.

This year Jim will buy 18 prizes.
He will spend the same total amount of money as he spent last year on the prizes.

Each of the 18 prizes must cost the same amount of money.

Work out the cost of each prize this year.

(3)

Step 1: Calculate total amount Jim spent last year

Total spent = 15 prizes × £27 each

15 × 20 = £300
15 × 7 = £105

Total = £405 [M1 - Total Amount]

Step 2: Cost of each prize this year (18 prizes for £405):

Cost per prize = £405 ÷ 18

405 ÷ 18 = 405 ÷ 9 = 45; 45 ÷ 2 = 22.5

= £22.50 [M1 A1]

- Check: 18 × £22.50 = 18 × 22 + 18 × 0.5 = 396 + 9 = £405 (Verified)
- Note: Money answers must always be given to 2 decimal places (£22.50, not £22.5).

£ **22.50**

(Total for Question 2 is 3 marks)

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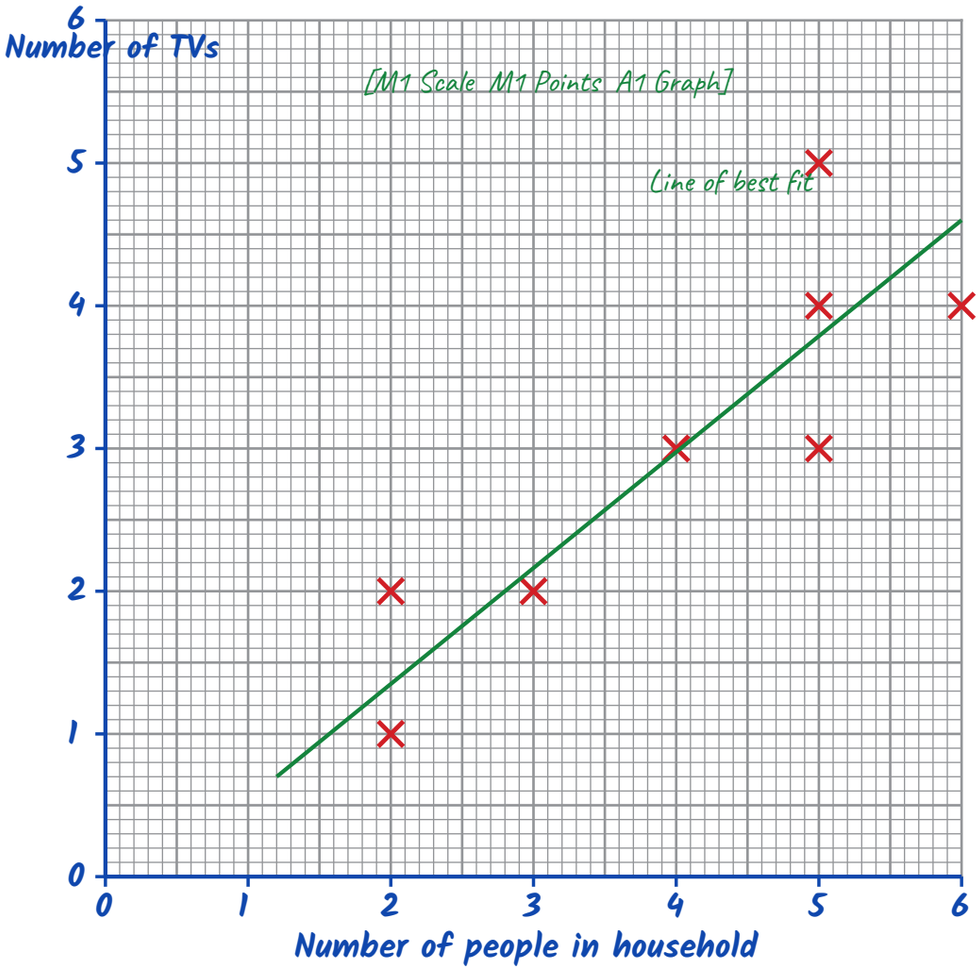
3 Steve is writing a report about TV viewing.
He has the following information.

Number of people in the household	3	5	5	6	5	2	4	2
Number of TVs in the household	2	3	4	4	5	1	3	2

Steve thinks there is a relationship between the number of people and the number of TVs in a household.

(a) Draw a suitable diagram for Steve.

(3)



(b) Describe the relationship between the number of people and the number of TVs in a household.

(1)

Positive correlation (As number of people increases, number of TVs increases)

[B1 - Correlation]

(Total for Question 3 is 4 marks)

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4 Karen invests £2 500 for 3 years.

The investment earns 1.7% compound interest per year.

Work out the value of the investment at the end of 3 years.

(3)

Step 1: Determine the annual compound interest multiplier

Interest rate = 1.7% per year

Multiplier = 100% + 1.7% = 101.7% = 1.017 *[M1 - Multiplier 1.017]*

Step 2: Calculate investment value over 3 years

Method A: Year-by-year calculation:

- *End of Year 1: £2500 × 1.017 = £2542.50*
- *End of Year 2: £2542.50 × 1.017 = £2585.7225*
- *End of Year 3: £2585.7225 × 1.017 = £2629.67978...*

Method B: Direct formula:

Value = £2500 × $(1.017)^3$ *[M1 - Full Process]*

= £2500 × 1.0518719... = £2629.67978...

Step 3: Round to 2 decimal places (pence):

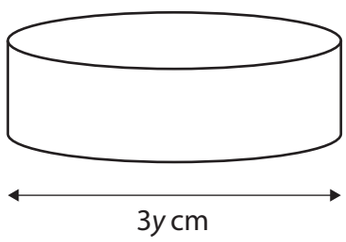
£2629.679... -> £2629.68 *[A1 - £2629.68]*

£ **2629.68**

(Total for Question 4 is 3 marks)

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5 Here is a cylinder.



h = 11 cm

diameter d = 3 × 11 = 33 cm

(a) Using $y = 11$, calculate the area of the curved surface of the cylinder.

(4)

Step 1: Find dimensions of cylinder for y = 11

- Height (h) = y = 11 cm
- Diameter (d) = 3y = 3 × 11 = 33 cm [M1 - Substitute y=11]
- Radius (r) = 33 ÷ 2 = 16.5 cm

Step 2: Curved surface area formula of a cylinder

Curved Area = Circumference × Height = π × d × h

(or formula: 2 × π × r × h)

Area = π × 33 × 11 [M1 M1 - Curved Area]

= 363 π

Step 3: Calculate numerical value

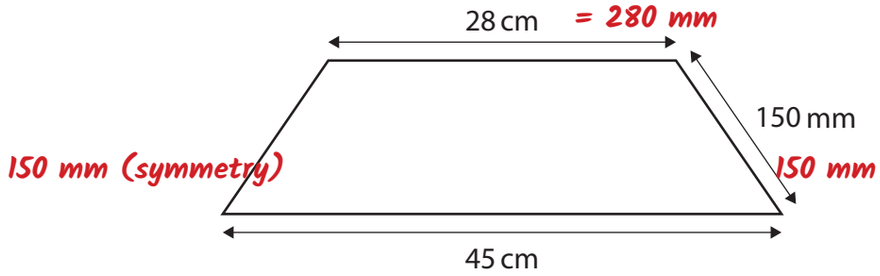
- Using π button on calculator: 363 × 3.14159... = 1140.398... cm²
- Using π = 3.14: 33 × 11 × 3.14 = 363 × 3.14 = 1139.82 cm²

= 1140.4 cm² (or 1140 cm²) [A1 - 1140 or 1139.82]

1140.4	cm^2

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Here is a trapezium.
The trapezium has one line of symmetry.



45 cm = 450 mm

(b) Choose the correct calculation to work out the perimeter of the trapezium.

(1)

- ☐ 28×150
- ☐ $28 + 150 + 45 + 150$
- ☐ 45×0.15
- ☑ [Correct Calculation] [B1]
- ☐ 280×150

Why this calculation is correct:

1. Perimeter is the total distance around the outside of the shape.
2. The shape has 1 line of symmetry (isosceles trapezium):
   the two slanted sides are equal (150 mm each).
3. All dimensions must be in consistent units before adding:
   - 28 cm = 280 mm
   - 45 cm = 450 mm
   - Slanted sides = 150 mm and 150 mm
4. Total perimeter = 280 + 150 + 450 + 150 = 1030 mm.

(Total for Question 5 is 5 marks)

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6

(a) Write 45% as a fraction in its simplest form.

(1)

Step 1: Write percentage as a fraction over 100:

$$45\% = \frac{45}{100} \begin{matrix} \div 5 \\ \div 5 \end{matrix} = \frac{9}{20}$$ *(divide by HCF = 5)* *[B1]*

9/20

(b) Calculate $\frac{5 + 13^2}{11 - 9.5}$

(2)

Step 1: Evaluate the Numerator using BIDMAS (Indices first):

- $13^2 = 13 \times 13 = 169$
- $5 + 169 = 174$ *[M1 - Numerator 174]*

Step 2: Evaluate the Denominator:

- $11 - 9.5 = 1.5$

Step 3: Divide numerator by denominator:

$174 \div 1.5 = 348 \div 3 = 116$ *[A1 - 116]*

116

(Total for Question 6 is 3 marks)

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7 Kate is making biscuits.
She uses flour, butter and sugar in the ratio 5 : 3 : 2

Kate has

- 800 g flour
- 550 g butter.

Kate wants to make the maximum number of biscuits possible.
She works out how much sugar she needs.

(a) How much sugar does Kate need?
You **must** show your working.

(3)

Ratio is Flour : Butter : Sugar = 5 : 3 : 2

Step 1: Find 1 part value to find limiting ingredient:

- **Flour: 800 g ÷ 5 parts = 160 g per part** *[M1 - Ratio parts]*
- **Butter: 550 g ÷ 3 parts = 183.33... g per part**

Step 2: Identify limiting ingredient and calculate sugar needed:

Since 160 < 183.33, Flour is the limiting ingredient.

Kate can make at most 160 batches/parts.

Sugar needed = 2 parts × 160 g = 320 g *[M1 A1]*

320 g

☑ (b) Use a reverse calculation to show a check of your working.

(1)

Reverse calculation to check 1 part value:

320 g ÷ 2 parts = 160 g per part *[B1 - Reverse Check]*

And 160 g × 5 parts = 800 g flour (matches available flour exactly - Verified)

320 ÷ 2 = 160

(Total for Question 7 is 4 marks)

8 Lou wants to go to a gym.

She sees these two offers at the gym.

Offer A	Offer B
Gym membership	Pay as you go
£44 per month + £30 joining fee	£7.75 per class
All classes included	No joining fee

Lou wants to attend 2 classes each week for one year.

Lou thinks if she uses offer A she will save at least 28% on the total cost of classes compared to using offer B.

Is she correct?
Show why you think this.

(6)

Step 1: Calculate total cost for Offer A (Gym membership for 1 year)

- 1 year = 12 months
- Monthly cost = 12 × £44 = £528
- Joining fee = £30

Total Offer A = £528 + £30 = £558 [M1 - Offer A = £558]

Step 2: Calculate total cost for Offer B (Pay as you go for 1 year)

- 1 year = 52 weeks
- Total classes = 2 classes/week × 52 weeks = 104 classes
- **Total Offer B = 104 × £7.75 = £806** [M1 - Offer B = £806]

Step 3: Calculate percentage saving with Offer A

- **Saving in pounds = £806 − £558 = £248**
- **Percentage saving = (£248 ÷ £806) × 100%**

= 30.769...% ≈ 30.8% [M1 M1 - % Saving]

Step 4: Comparison & Decision

30.8% is greater than 28% (30.77% ≥ 28%)

Yes, Lou is correct. [A1 - Decision Yes]

Decision: Yes, Lou is correct (Saves 30.8% >= 28%)

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Question 8 Alternative Method & Verification Summary

Alternative Method (Target Price with 28% Discount):

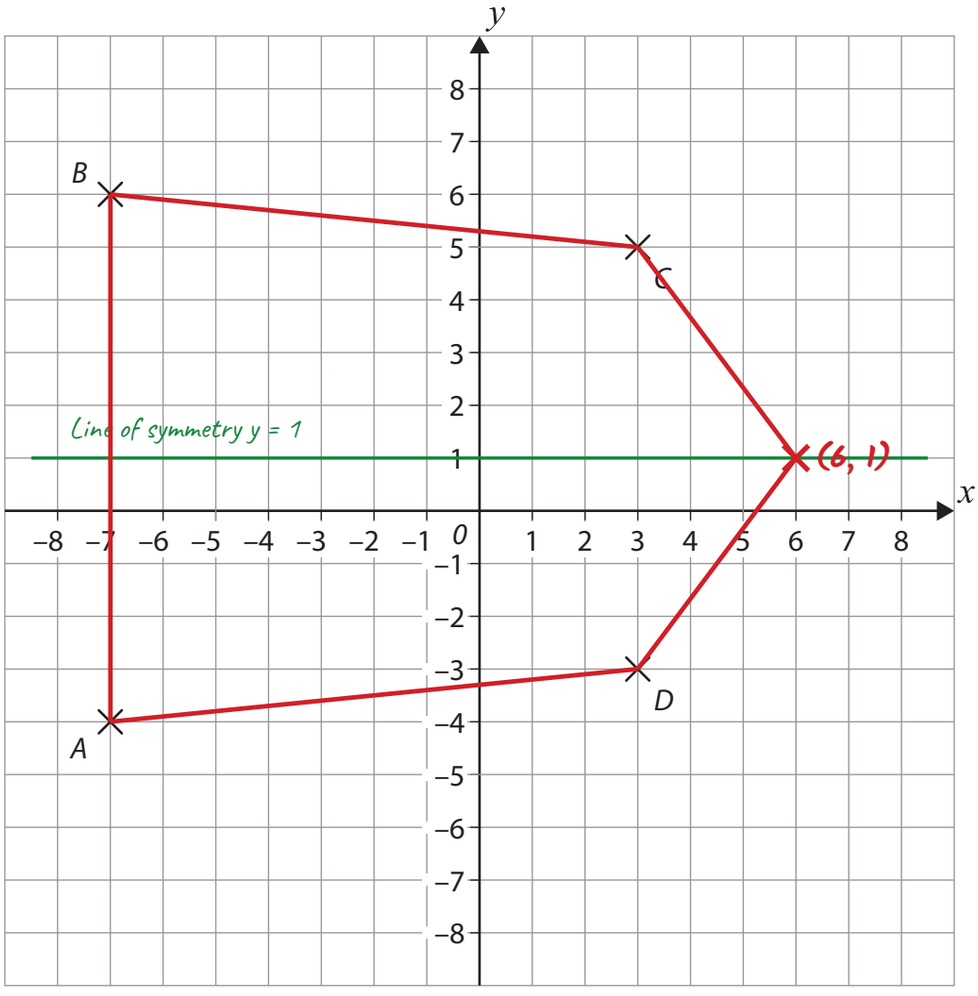
1. Offer B total cost = £806
2. 28% discount on Offer B = 0.28 × £806 = £225.68
3. Target maximum cost = £806 − £225.68 = £580.32
   (or: 0.72 × £806 = £580.32)
4. Actual cost with Offer A = £558
5. **Since £558 < £580.32, Lou saves MORE than 28%**
   -> Yes, Lou is correct (Confirmed)

Official Mark Scheme Breakdown (6 Marks):

- [M1] Process to calculate Offer A: 44 × 12 + 30 (= £558)
- [M1] Process to calculate Offer B: (7.75 × 2) × 52 (= £806)
- [A1] Both total figures accurate: £558 and £806
- [M1] Difference: 806 − 558 (= £248) OR 806 × 0.28 (= £225.68)
- [M1] Percentage saving: (248 ÷ 806) × 100 (= 30.77%)
- [A1] Valid decision "Yes" supported by accurate figures
  (30.8% or £580.32 vs £558)

(Total for Question 8 is 6 marks)

9 Here is a grid with points *A*, *B*, *C* and *D* plotted.



(a) Write down the coordinates of point *A*.

(1)

[B1 - Point A (-7, -4)]

(−7 , −4)

(b) On the grid, mark with a cross another point and join all five points to form a pentagon with one line of symmetry.

(2)

• The line of symmetry is y = 1 (midpoint of B and A, and C and D).

• 5th vertex lies on mirror line (e.g. at (6, 1)). 5 vertices form symmetric pentagon. [M1 A1]

(Total for Question 9 is 3 marks)

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10 Rav provides food for events.
Last year 550 people attended an event.
This year there will be 748 people attending the event.

Calculate the percentage increase in the number of people attending.

(3)

Step 1: Calculate the actual increase in number of people

$$\text{Increase} = 748 - 550$$

$$= 198 \text{ people}$$

[M1 - Increase = 198]

Step 2: Percentage increase on original attendance (550):

$$\text{Percentage Increase} = \frac{\text{Actual Increase}}{\text{Original Amount}} \times 100\%$$

$$= \frac{198}{550} \times 100\%$$

$$= 0.36 \times 100\%$$

$$= 36\%$$

[M1 A1]

Check with reverse multiplier:

$550 \times (1 + 0.36) = 550 \times 1.36 = 748$ people (checks out exactly - Verified)

36 %

(Total for Question 10 is 3 marks)

11 Meg has a dog.

She has this information about dog food.

Dog size	Daily quantity of dog food in packs
Small (up to 12 kg)	$\frac{1}{4}$ to $1\frac{1}{3}$
Medium (13 kg to 25 kg)	$1\frac{1}{3}$ to $2\frac{1}{3}$ <- Medium category
Large (26 kg to 45 kg)	$2\frac{1}{3}$ to $3\frac{3}{4}$
Giant (46 kg to 70 kg)	$3\frac{3}{4}$ to 5

Meg knows that her dog weighs 29 lbs.

1 kg = 2.2 lbs

Meg has 24 packs of dog food.

(a) What is the maximum number of days she can feed her dog with the 24 packs? Show why you think this.

(3)

Step 1: Convert dog weight from lbs to kg (1 kg = 2.2 lbs)

Weight in kg = 29 lbs ÷ 2.2 = 13.1818... kg [M1 - 13.18 kg]

Step 2: Identify dog size category from table

13.18 kg falls in the "Medium (13 kg to 25 kg)" category.

Daily feed range for Medium dog: 1 1/3 to 2 1/3 packs per day.

Step 3: Calculate the MAXIMUM number of days with 24 packs

To MAXIMISE the number of days, use the MINIMUM daily feed amount:

Minimum daily feed = 1 1/3 packs = 4/3 packs

Number of days = 24 ÷ (4/3) = 18 days [M1 A1]

18 days

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Meg needs to replace part of her garden fence to keep her dog safe.
She needs to replace 78 feet of fencing.

Meg buys fencing measured in metres.

Meg uses 1 metre is 3.28 feet.

(b) Work out how many metres of fencing Meg needs.

(2)

Step 1: Convert feet to metres (1 metre = 3.28 feet)

Metres of fencing = 78 feet ÷ 3.28 feet per metre

= 23.780487... metres *[M1 - 78 ÷ 3.28]*

Step 2: Functional rounding for practical purchasing:

- **To 2 decimal places: 23.78 metres** *[A1 - 23.78]*
- *(Pearson Mark Scheme also accepts 23.8 m or 24 m rounded up)*

Reverse check:

23.78 × 3.28 = 77.9984 ≈ 78 feet (Verified)

23.78 metres

(Total for Question 11 is 5 marks)

12 Ria works in a factory making solid steel spheres.

She has this information about the number of spheres made each day for the last 100 days.

Number of spheres	Number of days
210 to 212	4
213 to 215	32
216 to 218	53
219 to 221	11

Ria will use this information to estimate the mean number of spheres made each day.

She knows that the volume, v, of a sphere is

$$v = \frac{4\pi r^3}{3}$$

r is the radius of the sphere

Each sphere has a diameter of 15 mm.

Calculate the mean volume of steel used each day to make spheres.
Give your answer to the nearest thousand.

(6)

Number of spheres	Number of days	Midpoint	f × Midpoint
210 to 212	4	211	4 × 211 = 844
213 to 215	32	214	32 × 214 = 6848
216 to 218	53	217	53 × 217 = 11501
219 to 221	11	220	11 × 220 = 2420
	Total = 100		Total = 21613

Step 1: Calculate Estimated Mean number of spheres made per day

$$\text{Mean} = \frac{\text{Total } (f \times x)}{\text{Total } f} = \frac{21613}{100} = 216.13$$ [M1 M1 - Mean]

Step 2: Calculate Volume of 1 Steel Sphere

- Diameter = 15 mm -> Radius $r = 15 \div 2 = 7.5$ mm
- $r^3 = (7.5)^3 = 421.875$ mm^3
- **Volume = $(4/3)\pi(7.5)^3 \approx 1767.15$ mm^3** [M1 M1 - Volume]

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Step 3: Calculate Mean Volume of Steel Used Each Day

Mean Volume = (Mean number of spheres) × (Volume of 1 sphere)

= 216.13 spheres × 1767.14586... mm^3

= 381933.23... mm^3 [M1 - Daily Volume]

(Using π = 3.14: Volume = 1766.25 mm^3
-> 216.13 × 1766.25 = 381,739.6... mm^3)

Step 4: Round answer to nearest thousand (as specified):

• 381933 mm^3 rounded to the nearest thousand is:

382 000 mm^3 [A1 - Rounded]

Method Checklist & Examination Guidance:

* Find midpoints of grouped intervals: 211, 214, 217, 220
* Multiply each frequency by its midpoint and sum: Total(f × x) = 21613
* Divide by total frequency (100) to get estimated mean = 216.13 spheres
* Radius r = 7.5 mm; Volume = (4/3)π(7.5)³ = 1767.15 mm^3
* **Mean spheres × sphere vol = 381,933 mm^3 -> 382 000 mm^3**

382 000 mm^3

(Total for Question 12 is 6 marks)

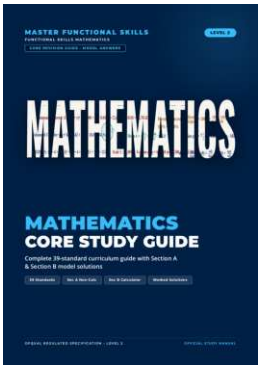
TOTAL FOR SECTION B IS 48 MARKS
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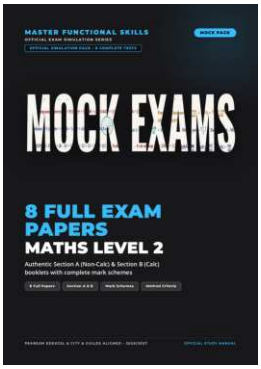


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